

Assignment 3 – Clustered Forward Shading

Advanced Computer Graphics, 5DV180

Olle Lögdahl
olle.logdahl@umu.se

I. INTRODUCTION

The following document describes the design of a clustered forward shading renderer, backed by GPU-driven rendering. The new renderer supports many lights (up to 10'000 in real time).

II. USAGE GUIDE

In the following section, the usage of the program is described. Firstly in Section II.A, the building process is described.

A. Building and running

For building the application, the following are required:

- C++20 / C++2a compatible compiler.
- GLFW installed system-wide.
- make or cmake (and a buildsystem like make or ninja)

The following make commands are available:

- `make dev` (default) - Builds the program in dev mode. Full optimizations but with tracing enabled.
- `make release` - Builds the program in release mode.
- `make memcheck` - Builds the program with ubsan, asan and validation layers enabled.
- `make clean` - Cleans the build directory.

After building the program, it can be run with `./cgr <scene file>`. The program requires `glslc` [1] to be installed and on `$PATH`, as it is used to compile glsl shaders. If `glslc` is not found, the path can be specified using the `GLSLC_PATH` environment variable at runtime.

The project can also be built using cmake:

- `cmake -B build -DCMAKE_BUILD_TYPE=Debug`
- `cmake --build build --parallel`

B. System Requirements

Below are the system requirements for running the application. Note that the program only works on linux currently.

- Vulkan 1.3 compatible GPU. The descriptor indexing feature is also required. These are generally widely supported.

III. BACKGROUND

A classical forward renderer suffers from some issues, the biggest usually being overdraw. An overdrawn fragment is both wasted work, as well as inefficient usage of fill rate. As

each fragment is being rendered in a forward-fashion, costly lighting calculation occurs for fragments which will not be seen. Historically, deferred rendering has been proposed to solve this. By rendering only albedo, normals and other properties per-object, and later shading by pixel, the shading is only performed on every visible fragment. While this solves overdraw, deferred shading comes with other issues relating to bandwidth as well as anti-aliasing.

The issue of overdraw can be handled in many ways, like performing an early depth-only pass or by triangle reordering. [2] In this assignment, we focused on lessening the overdraw impact by reducing the cost of fragment shading. Our method, Clustered Shading, can also efficiently be applied to a deferred renderer to support many lights. Our solution is capable of rendering a scene with 10'000 point lights in real time.

IV. CLUSTERED FORWARD SHADING

Clustered shading is a method to reduce lighting calculations by reducing the number of lights that are tested against each fragment. [3] The method works by splitting the view frustum into frustum shaped boxes called clusters and assigning the lights to the clusters based on their influence radius. When shading, each fragment can look up the containing cluster and only calculate shading using those lights. This means that clusters with no lights perform no lighting calculations.

In our implementation, we define the clusters using axis-aligned bounding boxes in view space. This is not completely accurate, as there will be slight overlap, but this doesn't cause any issues in the tested scenes.

A great feature of clustering is that the system can be reused for other spatial-related systems like decals and light probes. [4]

A. Cluster Generation

The first step is to generate the clusters bounding boxes. The clusters needs to be recalculated every time the projection changes. For clustering, we define a grid size for the number of clusters in the frustum; we use the same as [4] which is 16x8x24. Each compute invocation calculates the bounding box for a single cluster, using its invocation ID to calculate the cluster position in view space.

While the X and Y axis are divided into equal parts, the Z axis is sliced logarithmically to counteract both the fact that the NDC space is not linear and that clusters closer to the camera

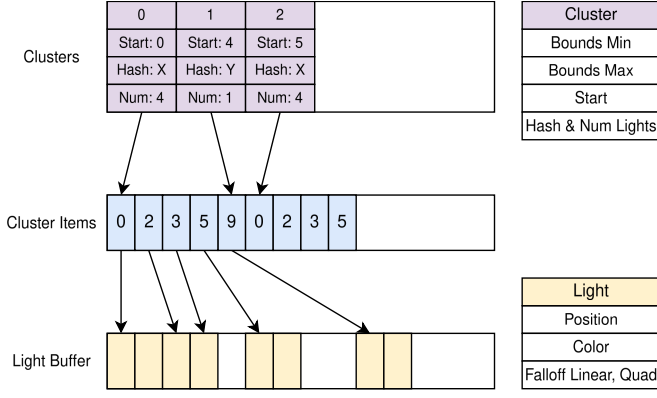


Figure 1: Memory layout of clusters, cluster items and lights.

likely cover more fragments. The start depth of slice n (Z_n) is calculated using (1), where N is the total number of Z slices. [4]

$$Z_n = Z_{\text{near}} * \left(\frac{Z_{\text{far}}}{Z_{\text{near}}} \right)^{\frac{n}{N}} \quad (1)$$

This gives us a cheap mapping between view-space position and cluster index.

B. Cluster Assignment

Assigning lights to clusters is the main part of the clustering algorithm. In previous assignments we already store all point lights in a buffer, so we use an indirection buffer similar to [4] to reference them. After this pass, each cluster will have a reference to the start of the cluster items, as well as a hash and the number of lights. See Figure 1 for a memory layout of clusters, cluster items and lights.

Using a naive approach, we were able to assign 10'000 lights to 2'800 clusters in 6ms on a GTX 1070. While an extreme case, we optimized the algorithm to perform faster.

C. Forward Shading

D. Optimized Scalar Reads

V. LIMITATIONS

A. Reflections

The point lights influence is calculated using the range of diffuse lighting. Specular lighting has no upper range, and would require an influence range of infinity. This means that normal specular highlights no longer work. This means that reflections need to be handled in another way.

Luckily, reflections calculated using the lights is already very limited. Metallic surfaces usually become very dark, as there is rarely anything to reflect. This means that most engines already use another method for reflections. We suggest the following methods:

- Ray-traced reflections
- Screen space reflections
- IBL Cubemap reflections

Each method has its own advantages and disadvantages. Ray-traced reflections are the most accurate, but also likely the

most expensive. Screen space reflections are cheaper, but have a problem with things outside the screen, requiring another method to fall back on. IBL Cubemap reflections are a good idea, as it synergizes well with the current clustering system, but has some PBR issues.

Cubemap reflections, hereforth called *Reflection Probes*, synergizes well with our clustering approach. Each cluster could track both lights and reflection probes, meaning that local reflections could be possible. One problem with reflection probes is that fragments not on the probe origin will be reflected incorrectly. This can be compensated for using parallax correction, but this doesn't fully solve it.

BIBLIOGRAPHY

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- [4] T. Sousa, and J. Geffroy, "The devil is in the details. Presented at Advances Real-Time Rendering Games Course, 2016.